

BAB V

PENUTUP

5.1. Kesimpulan

Setelah dilakukan langkah-langkah penelitian yang ada, didapatkan hasil dalam mengimplementasikan sistem otomatisasi berbasis *visual programming* untuk budidaya basil hidroponik, maka dapat diambil beberapa kesimpulan, diantaranya:

1. Implementasi *visual programming* untuk otomatisasi budidaya basil hidroponik berhasil dilakukan dengan mengubah arsitektur kendali konvensional menjadi terpusat pada server. Sistem ini dibangun menggunakan perangkat lunak N8N sebagai pusat logika, MQTT Broker sebagai jalur komunikasi, dan MariaDB sebagai penyimpanan data. Pada sisi perangkat keras, sistem diimplementasikan menggunakan *microcontroller* ESP32, modul sensor analog (*pH* dan TDS), sensor suhu DS18B20, sensor ultrasonik, modul *relay*, serta rangkaian pompa mini dan *solenoid valve* yang dikemas dalam panel kontrol terintegrasi.
2. Berdasarkan hasil pengujian sistem yang dilakukan secara kontinu selama 72 jam (3 hari), sistem terbukti mampu beroperasi secara mandiri (*autonomous*) dalam memantau dan mengoreksi kondisi air. Dari evaluasi akurasi, diperoleh nilai *Mean Absolute Error (MAE) pH* sebesar 0,72 dan *MAE TDS* sebesar 7,82 ppm. Nilai penyimpangan yang rendah ini menunjukkan bahwa sistem memiliki tingkat presisi yang baik. Mekanisme logika kontrol juga terbukti berfungsi efektif dalam mengembalikan parameter air ke rentang ideal dengan rata-rata waktu proses $\pm 3-5$ menit.

5.2. Saran Pengembangan

Berdasarkan hasil analisis, kendala teknis, dan batasan masalah yang ada pada penelitian ini, peneliti mengajukan beberapa saran untuk pengembangan sistem di masa mendatang, diantaranya:

1. Disarankan untuk mengganti modul sensor TDS dan *pH* analog dengan sensor tipe *industrial grade* yang memiliki material *probe* lebih tahan korosi dan fitur isolasi elektrik internal. Hal ini bertujuan untuk menjaga stabilitas

pembacaan dalam penggunaan jangka panjang serta mengurangi frekuensi perawatan atau kalibrasi ulang dibandingkan dengan penggunaan sensor standar.

2. Pengembangan selanjutnya dapat memperluas logika sistem yang saat ini masih berbasis aturan (*rule-based*) dengan mengintegrasikan *Machine Learning* pada *workflow* N8N. Implementasi ini memungkinkan sistem untuk memprediksi kebutuhan nutrisi tanaman berdasarkan tren data historis, bukan hanya bereaksi terhadap kondisi saat ini.
3. Penelitian selanjutnya dapat mengembangkan cakupan otomatisasi tidak hanya pada parameter larutan nutrisi, tetapi juga mencakup aspek lingkungan lainnya yang mempengaruhi pertumbuhan tanaman basil, seperti pengendalian intensitas cahaya dan sirkulasi udara berdasarkan kelembaban ruang.
4. Disarankan untuk memperpanjang durasi pengujian sistem secara berkesinambungan selama 1 hingga 3 bulan. Mengingat pengujian pada penelitian ini hanya dilakukan selama 72 jam (3 hari), pengujian jangka panjang diperlukan untuk mengevaluasi ketahanan perangkat keras, stabilitas kinerja sistem otomatis dari waktu ke waktu, serta dampak langsung intervensi sistem terhadap kualitas pertumbuhan tanaman.

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